Tan-Activity-1

2026-09-28

Main research interest

My main research interests include remote sensing, forest and vegetation ecosystem monitoring, aboveground biomass and carbon dynamics, and the application of machine learning/deep learning to remote sensing. **My previous research** has focused on forest tree species mapping, biomass estimation, and forest disturbance monitoring. These studies were mainly conducted over relatively limited spatial extents. **Currently**, I am interested in extending my research to larger **regional/global scales** to understand the dynamics and functioning of forest and vegetation ecosystems. **A particular focus of my research** is understanding how terrestrial ecosystems respond to global warming and other environmental changes, and how these changes affect ecosystem carbon cycling. I hope to use long-term satellite observations, multi-source remote sensing data, and machine learning approaches to monitor changes in vegetation productivity, biomass, carbon stocks, and ecosystem disturbances at large spatial and temporal scales. I also aim to examine how factors such as temperature, drought, wildfire, land-use change, and other disturbances influence ecosystem carbon dynamics and resilience.

Favorites

Favorite Music

1. "Merry Christmas Mr.Lawrence"- Ryuichi Sakamoto
2. "Imagine"- John Lennon
3. "Bohemian Rhapsody"- Queen
4. "The Great Gig in the Sky"- Pink Floyd
5. "Untitled(How Does It Feel)"- D'Angelo

Favorite Equation

Newton–Leibniz Formula

$$\int_a^b f(x)\,dx = F(b) - F(a), \qquad \text{where } F'(x) = f(x)$$

Favorite Artists

Name	Achievements
Christopher Nolan	Director, Academy Award winner
Hans Zimmer	Composer, Academy Award winner
Lang Lang	Pianist, International concert performer
Ryuichi Sakamoto	Composer and pianist, Academy Award winner

Name	Achievements
Ieoh Ming Pei	Architect, Pritzker Prize winner

A Chunk of Code

```
# Randomly two limits between 0-20
limits <- sort(sample(0:20, 2))

lower <- limits[1]
upper <- limits[2]

# Define f(x)= x^2
f <- function(x) {
  x^2
}

# Calculate
result <- integrate(f, lower = lower, upper = upper)

# print limits and integral value
cat("Lower limit:", lower, "\n")
```

```
## Lower limit: 8
```

```
cat("Upper limit:", upper, "\n")
```

```
## Upper limit: 18
```

```
cat("Integral value:", result$value, "\n")
```

```
## Integral value: 1773.333
```